10-Year Financial Asset Analysis

Bitcoin · Gold · Crude Oil · US Dollar Index (DXY)

Analysis Period: January 2015 – January 2026

DATA SCIENCE FINAL PROJECT

BTC | GOLD | OIL | DXY

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Executive Summary

This report presents the findings of a comprehensive, data-driven analysis of four major financial assets — **Bitcoin (BTC), Gold (XAU), Crude Oil (WTI)**, and the **US Dollar Index (DXY)** — over a ten-year period spanning January 2015 to January 2026. The study was conducted through a structured 13-stage analytical pipeline encompassing data cleaning, return analysis, normalization, correlation mapping, outlier detection, statistical hypothesis testing, and advanced visualization. The objective is not merely to describe historical price action, but to extract actionable structural insights about the relationships, risks, and behavioral dynamics of these assets — and to frame those insights within a forward-looking analytical context.

The decade under review was anything but ordinary. It encompassed the birth of institutional Bitcoin adoption, a global pandemic, a historic oil price collapse, aggressive monetary tightening, geopolitical conflict in Europe, and the emergence of crypto as a recognized asset class. Each of these events left measurable fingerprints in the data — and this analysis maps them precisely.

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Key Findings: Bitcoin & Gold

Bitcoin (BTC)

Bitcoin delivered the most extraordinary return profile of the four assets — growing from approximately $300 in early 2015 to over $100,000 by late 2024, representing a multi-thousand-percent cumulative gain. However, this performance came paired with the highest daily volatility (standard deviation exceeding 3% per day), the deepest maximum drawdowns (exceeding 80% on multiple occasions), and the most outlier events of any asset in the study.

Bitcoin showed strong positive performance in 2017, 2020, 2021, and 2024, but suffered catastrophic declines of over 70% in 2018 and over 65% in 2022. Despite the extreme risk profile, Bitcoin's Sharpe Ratio remained competitive — suggesting that its excess returns have historically compensated investors who held through the volatility. The asset displays a clear **four-year halving cycle pattern**, with major bull markets following each supply reduction event in 2017, 2021, and 2024.

Gold (XAU)

Gold fulfilled its traditional role as a stable store of value and safe-haven asset throughout the decade. It delivered steady, compounding appreciation — rising from approximately $1,200 per ounce in 2015 to near $2,600–$2,700 by 2025 — while maintaining significantly lower volatility than both Bitcoin and Oil.

Gold's most notable surge occurred in 2020, when global uncertainty from the COVID-19 pandemic drove capital toward safety. It also demonstrated resilience in 2022 and 2023 amid geopolitical stress from the Russia-Ukraine conflict. Gold's correlation with DXY is consistently inverse — a fundamental macroeconomic relationship that held robustly across the entire study period. **Gold is the most predictable and structurally stable asset in this study.**

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Key Findings: Crude Oil & DXY

Crude Oil (WTI)

Oil was the most externally disrupted asset in the study. It experienced the most dramatic single event of the entire dataset: in April 2020, WTI crude oil futures briefly traded at **negative prices** — an unprecedented event driven by the collapse of global demand during COVID-19 lockdowns combined with a storage capacity crisis.

Beyond this historic event, Oil demonstrated a recovery and rally cycle from mid-2020 through 2022, partially driven by the geopolitical shock of the Russia-Ukraine conflict reducing global supply. Oil's correlations with other assets are the weakest of the group, making it the **most idiosyncratic asset**. Its price is heavily governed by OPEC+ production decisions, geopolitical events in producing regions, and seasonal demand patterns — factors that are largely exogenous to financial market dynamics.

US Dollar Index (DXY)

DXY functions as a **macro stability barometer** rather than a return-generating investment. Over the decade, it oscillated within a relatively bounded range of approximately 88 to 115 units, with the most significant move being the aggressive 2022 rally driven by the Federal Reserve's historic interest rate hiking cycle.

DXY maintains a consistent inverse relationship with Gold — one of the most stable macro correlations confirmed in this study. It also shows negative correlation with Bitcoin during periods of risk-off sentiment. For portfolio construction, DXY (or dollar-denominated instruments) provides the most effective diversification hedge against risk assets during periods of financial stress.

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Correlations & Inter-Asset Relationships

One of the most important findings of this study concerns the **time-varying and regime-dependent nature of correlations** between assets. At the daily return level, correlations between all four assets are generally low to moderate — which is theoretically favorable for diversification. However, the rolling 60-day correlation analysis reveals a critical phenomenon: during periods of acute market stress, correlations between risk assets tend to converge upward simultaneously. This is known as **Correlation Convergence**, and it represents the single greatest challenge to the naive diversification thesis — assets that appear uncorrelated in normal conditions begin moving together precisely when diversification is needed most.

Gold ↔️ DXY

Strong, consistent inverse relationship. When the dollar strengthens, gold faces headwinds — and vice versa. This is the **most stable macro correlation** in the dataset.

Bitcoin ↔️ Gold

Moderate positive correlation, especially during crisis periods, as both attract capital as inflation hedges and alternative stores of value.

Oil ↔️ DXY

Moderate inverse relationship. A stronger dollar makes dollar-denominated oil more expensive for foreign buyers, reducing demand.

Bitcoin ↔️ Oil

Weak correlation in normal conditions, but both can react sharply to macro risk-off events.

Monthly vs. Daily Correlations

Correlations are consistently stronger at the monthly timeframe than at the daily timeframe, suggesting that short-term noise masks longer-term structural relationships.

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Statistical Analysis Findings

The statistical testing stage produced several findings with direct implications for how these assets should be modeled and analyzed:

Test	Result	Implication
ANOVA (Return Comparison)	Null hypothesis rejected ($p < 0.05$)	Statistically significant differences exist between the return profiles of these assets
Shapiro-Wilk Normality Test	Rejected for all four assets	None of the assets follow a normal distribution
Jarque-Bera Test	Rejected for all four assets	Fat tails and skewness confirmed — extreme events occur far more frequently than standard models predict
Levene / Bartlett Test	Heteroscedasticity confirmed	Variance is not constant over time; GARCH-family models far more appropriate than constant-variance approaches
Mann-Whitney U (Post-Hoc)	Bitcoin significantly different from all others	Bitcoin occupies a distinct return regime
Hierarchical Clustering	BTC & Gold cluster together; DXY separate	Consistent with DXY's role as a counter-cyclical reserve currency asset

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Major Market Events — A Decade of Disruption

The outlier analysis systematically identified the largest single-day price moves across all four assets and mapped them to historical events. The following captures the most significant market-moving events of the decade:

2017 — Bitcoin Bull Run & Crypto Mania

Bitcoin surged from approximately $1,000 to nearly $20,000 in a single year, driven by retail speculation, ICO mania, and first-wave institutional interest. Gold was stable; Oil and DXY were largely unaffected.

2018 — Crypto Crash & Dollar Strength

Bitcoin collapsed over 80% as regulatory fears and speculative excesses unwound. The US dollar strengthened throughout the year as the Fed raised rates, putting sustained pressure on Gold.

March 2020 — COVID-19 Pandemic Shock

The single most impactful event of the decade. All risk assets sold off simultaneously in a liquidity crisis. Crude oil briefly went negative in April 2020 — an unprecedented event in commodity history. Bitcoin dropped approximately 50% in two days. Gold initially fell, then surged to record highs as the scale of monetary stimulus became clear.

2021 — Post-COVID Reflation & Crypto Peak

Unprecedented fiscal stimulus and near-zero interest rates created a perfect environment for risk assets. Bitcoin reached $69,000; Gold held near record highs; Oil recovered strongly. The Federal Reserve's November 2021 tapering announcement triggered the beginning of Bitcoin's sustained decline.

2022 — Rate Hike Shock & Geopolitical Crisis

The Federal Reserve's most aggressive rate hiking cycle since 1980 crushed risk assets across the board. Bitcoin fell over 65%. The Russia-Ukraine war in February 2022 caused Oil to surge past $130 per barrel and Gold to rally sharply. DXY reached its strongest level in 20 years. The FTX exchange collapse in November 2022 added a crypto-specific shock, erasing tens of billions in market capitalization within days.

2023–2024 — Bitcoin ETF Era & New All-Time Highs

The approval of spot Bitcoin ETFs in the United States in January 2024 marked a structural shift in institutional accessibility. Combined with the April 2024 halving, Bitcoin surpassed $100,000 for the first time. Gold also hit new all-time highs in 2024, driven by central bank buying and persistent geopolitical uncertainty. Oil remained volatile but range-bound.

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Forward-Looking Analysis & Predictive Outlook

The following observations are analytical projections based on identified cyclical patterns, macro correlations, and behavioral regimes — not financial advice.

Bitcoin: The Halving Cycle Continues

Bitcoin's historical behavior has followed a remarkably consistent four-year cycle tied to its programmatic supply halving events. The April 2024 halving reduced the block reward from 6.25 to 3.125 BTC. Based on the pattern established in previous cycles, this suggests a potential peak window in the 2025–2026 timeframe. However, each cycle also brings deeper institutional participation, which may compress volatility over time. The introduction of spot Bitcoin ETFs represents a structural shift in demand dynamics — new capital from pension funds, sovereign wealth funds, and retail investors through regulated vehicles creates a more persistent demand floor. Key risk factors to monitor include US regulatory developments, macroeconomic policy pivots, and on-chain metrics such as exchange reserves and miner activity.

Gold: Structural Tailwinds in a Deglobalization Era

Gold enters the post-2025 period with unusually strong structural tailwinds. Central bank gold buying has reached multi-decade highs, particularly from non-Western nations seeking to reduce dollar reserve dependency following the 2022 SWIFT sanctions on Russia. This represents a fundamental, non-speculative demand driver that is independent of investor sentiment cycles. Additionally, rising global debt levels and the long-term trajectory of real interest rates favor gold as a store of value. If the Federal Reserve continues its rate-cutting cycle — which data from 2024 began to confirm — gold historically outperforms during such periods. The DXY-Gold inverse relationship implies that any meaningful dollar weakening cycle would amplify gold's upside. A sustained move toward $3,000+ per ounce in the 2025–2026 window appears structurally supported.

Crude Oil: Supply Constraints vs. Energy Transition

Oil faces the most complex forward outlook of the four assets. On the supply side, OPEC+ production discipline has supported prices, while geopolitical risks in the Middle East and Russia continue to create upside volatility. On the demand side, the accelerating energy transition — particularly the rapid adoption of electric vehicles in China and Europe — creates a credible long-term demand ceiling. The data suggests that Oil is likely to remain range-bound and event-driven rather than trending strongly in either direction over the medium term. For analytical purposes, Oil is best modeled as a mean-reverting commodity with fat-tailed event risk rather than a trending financial asset.

DXY: The Interest Rate Cycle Defines Everything

The trajectory of DXY over the next two to three years will be largely determined by the Federal Reserve's interest rate policy path. The aggressive 2022–2023 hiking cycle produced the strongest dollar environment in 20 years. As the Fed transitions toward a cutting cycle — already initiated in late 2024 — DXY faces structural headwinds. A weaker dollar environment is historically positive for Gold, supportive for commodity prices including Oil, and generally risk-positive for Bitcoin. Monitoring the Fed Funds Rate, US inflation data, and the yield differential between the US and other major economies will be critical for projecting DXY direction and its ripple effects across all other assets in this study.

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Portfolio Construction Insights

Diversification Works — But Is Weakest When Needed Most

The four assets have sufficiently low average correlations to provide meaningful diversification benefits in a combined portfolio. However, rolling correlation analysis shows that during acute crises, these correlations spike simultaneously — reducing diversification exactly when protection is most valuable. Tail-risk hedging strategies such as options and volatility instruments are needed to address this structural gap.

Bitcoin as a High-Conviction Satellite Position

Given its extreme return potential and extreme risk profile, Bitcoin is best positioned as a satellite allocation — typically 1–10% of a portfolio — rather than a core holding. Its inclusion has historically improved the Sharpe Ratio of diversified portfolios, but position sizing must be carefully calibrated to an investor's actual drawdown tolerance.

Gold as the Anchor of the Alternatives Sleeve

Gold's combination of moderate returns, low correlation with equities, and safe-haven properties during crises makes it the most reliable alternative asset in this study. A 5–15% strategic allocation to Gold has historically improved risk-adjusted portfolio performance across a range of market environments.

Oil as a Tactical Rather Than Strategic Position

Oil's idiosyncratic, event-driven price behavior and limited long-term return profile make it better suited to tactical allocation — responding to supply disruptions and geopolitical events — than permanent strategic allocation in a long-term portfolio.

DXY (or Cash and Short-Term Bonds) as the Crisis Hedge

During risk-off environments, DXY strengthens while risk assets fall. Holding a portion of the portfolio in dollar-denominated short-duration assets provides the most effective crisis buffer available in this asset universe — and requires no active management to deliver its protective function.

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Conclusion

This ten-year analysis of Bitcoin, Gold, Crude Oil, and DXY reveals a world of deeply interconnected yet behaviorally distinct financial assets. The decade from 2015 to 2026 was defined by unprecedented monetary experiments, a global pandemic, geopolitical realignment, and the maturation of a new asset class. Each of these forces left measurable, quantifiable marks in the data — and this project has mapped them with precision.

> **No single asset dominates across all dimensions.** Bitcoin wins on return. Gold wins on stability. Oil wins on commodity utility. DXY wins on crisis resilience. The sophisticated investor does not choose one — they build a framework that leverages each asset's strengths at the right time, in the right proportion, within a clearly defined risk management structure.

As we look forward from 2026, the structural forces identified in this study — the Bitcoin halving cycle, gold's central-bank demand tailwind, oil's energy-transition headwind, and DXY's rate-cycle sensitivity — provide a coherent analytical framework for navigating whatever comes next. Markets will remain complex, non-linear, and driven by events that no model can perfectly anticipate. But disciplined, data-driven analysis of the kind demonstrated in this project is the most reliable compass available.

Methodology: This analysis employed a rigorous 13-stage data science pipeline: raw data ingestion and cleaning, datetime standardization, inner-join merging on trading dates, return computation, Min-Max normalization, cumulative performance analysis, pairwise scatter and regression analysis, monthly, quarterly, and annual resampling, rolling statistics, IQR-based outlier detection, multi-timeframe correlation analysis, hierarchical clustering, event annotation, and comprehensive statistical hypothesis testing covering ANOVA, Mann-Whitney U, Shapiro-Wilk, Jarque-Bera, Levene, and Kolmogorov-Smirnov.

Limitations: The analysis uses daily closing prices, which do not capture intraday volatility; the inner join approach means days where any single asset has missing data are excluded, which particularly affects weekends for traditional assets versus the 24/7 crypto market; all correlations and statistical findings reflect the specific 2015–2026 window and should not be extrapolated as permanent or universal relationships; and this analysis does not constitute financial advice — all forward-looking statements are analytical projections based on historical patterns.

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